

Brainwide Blood Volume Reflects Opposing Neural Populations

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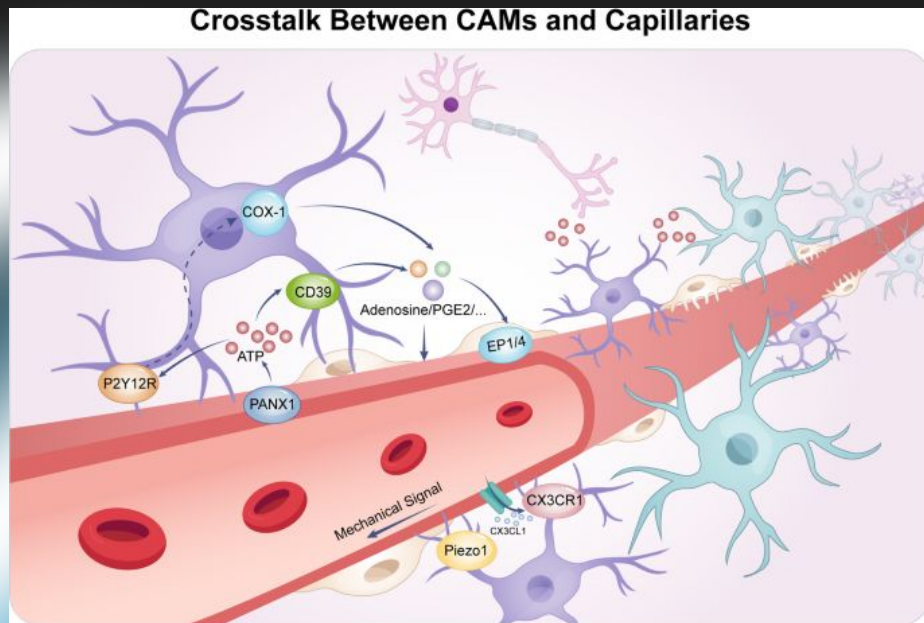
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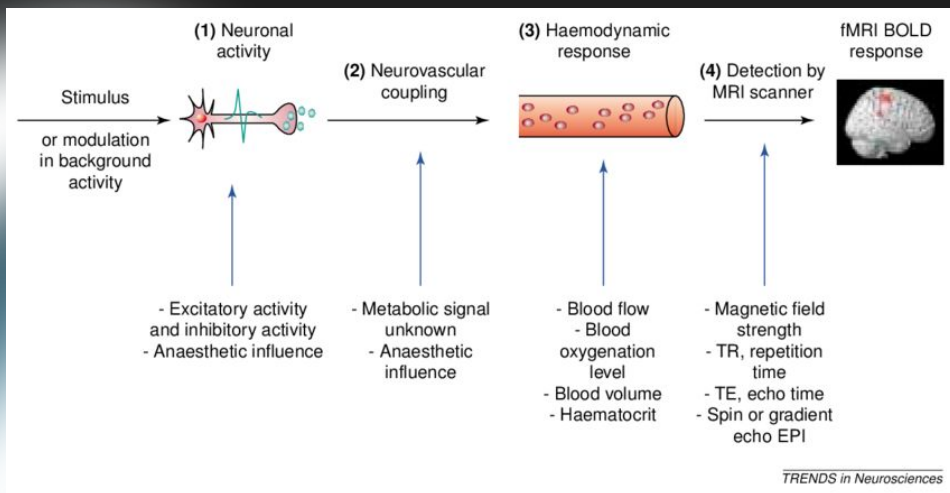
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Neurovascular Coupling



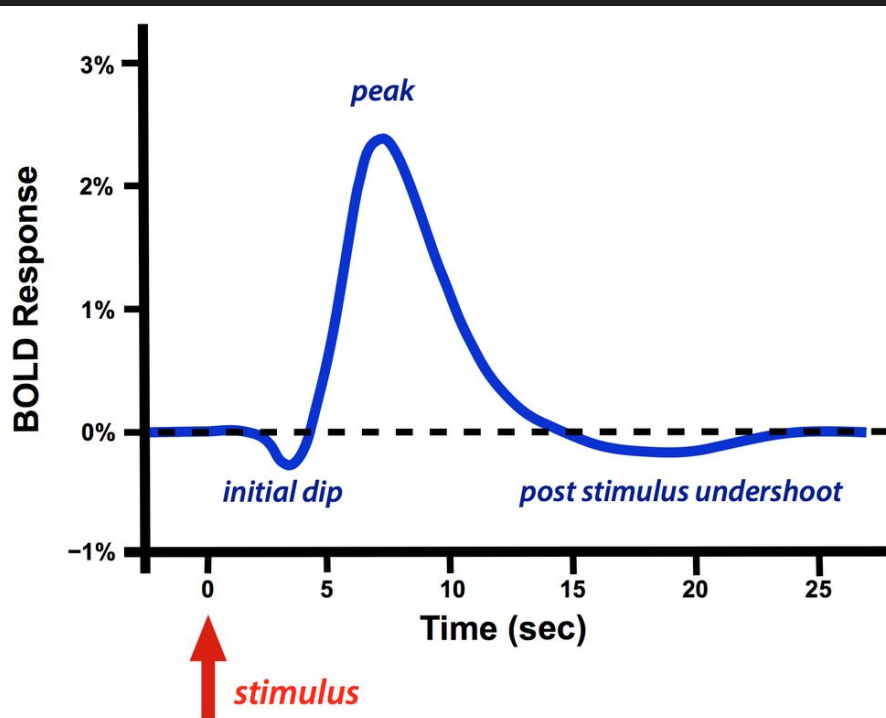
- When a region of neurons begin firing rapidly, fuel is required
- A fuel signal triggers widening of endothelial cells, bringing in oxygen and glucose
- Microglia (CAMs) act as sensors for such events

BOLD Signal



- Blood oxygen level dependent signal
- Primary metric used in fMRI to map brain activity
- Measures changes in blood flow and oxygenation in the brain

Hemodynamic Response Function HRF

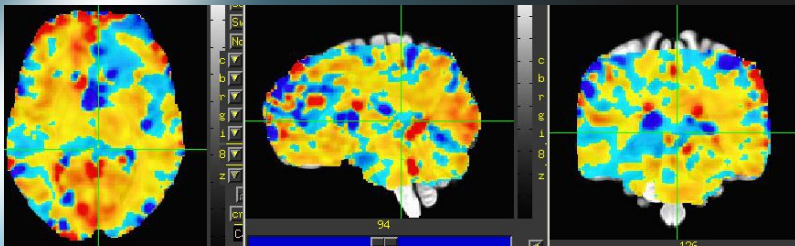


- A mathematical model that describes the process how blood supply changes in response to neural activity
- After a stimulus, there is an initial dip reflecting oxygen consumption, then a peak, and finally an undershoot during vascular recovery.

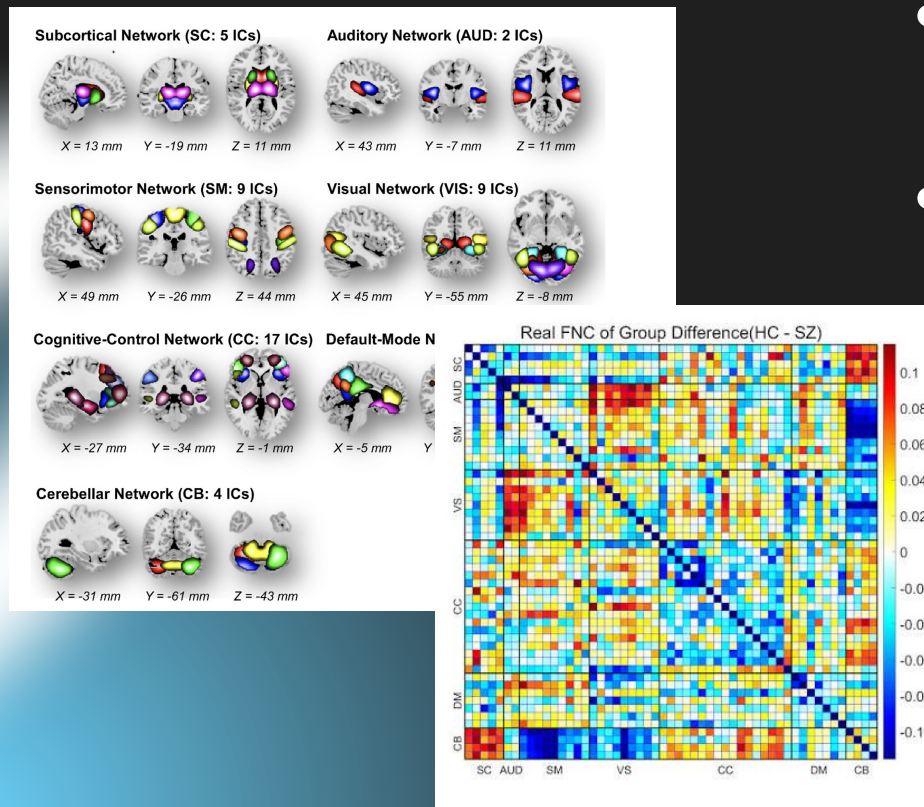
Fitting to the HRF

Design Matrix (X)			Real BOLD (Y)	
[Predicted	Baseline]		[Voxel Time-Series]	
[0.00	1.0]		[502.1	]
[0.01	1.0]		[501.8	]
[0.12	1.0]	vs	[505.4	]
[0.85	1.0]		[512.3	]
...			...	
[300 rows x 2 cols	]		[300 rows x 1 col	]

- A task vector is designed with impulse signals
 - on during task, off otherwise
- The HRF is evaluated at the sampling rate of the scanner
- This is convolved with the task vector, giving the predicted bold shape vector (X)
- A regression tries to find the slope for a line through the true BOLD signal (Y) and predicted (X)
- The slope of the line represents the magnitude of vascular response to the task. (per voxel)

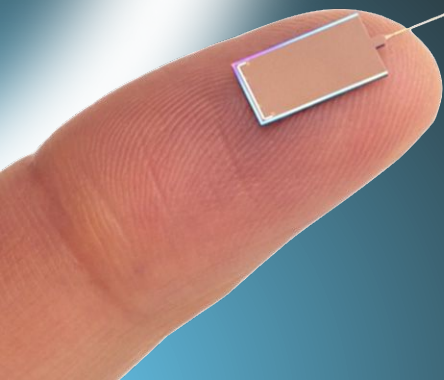


Connectivity in fMRI



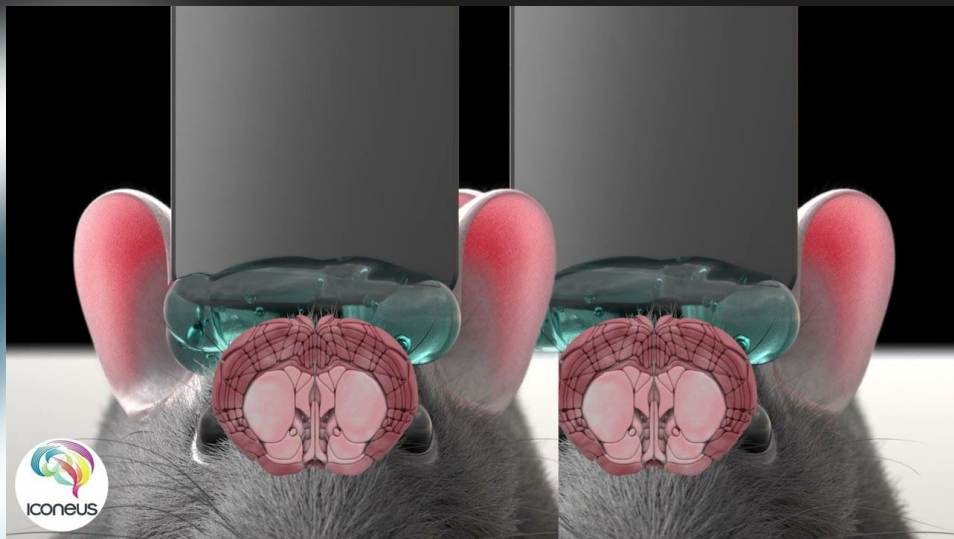
- Correlates BOLD timeseries between brain regions
- Positive correlation is often associated with activity/synchronicity between regions
 - Shared functional network
- Negative correlation is often associated with anticorrelation between regions
 - Competing cognitive states

Neuropixels



- Silicon based microelectrode probes designed to record the electrical activity of hundreds of individual neurons simultaneously with single cell resolution
- They have an ultra fast resolution and are able to capture exact electrical firing.

Functional Ultrasound Imaging fUSI



- An imaging technique based on the physics of sound waves to track moving blood cells in the brain.
- Uses the doppler effect to calculate frequency shifts in moving blood cells to build a realtime map of cerebral blood volume, a blood volume timeseries.

Whisking in Mice



Rhythmic back-and-forth movement of a mouse's facial whiskers.

Allows mice to map their environment through touch, sound, vibrations, etc.

When alert, aroused, or curious, a mouse's brain releases neuromodulators, triggering whisking.

Experiments Outline

1. Tracking Spontaneous Arousal
2. Classifying Neurons by Arousal Response
3. Testing Across Sleep and Wake
4. Mapping Population Distribution
5. Predicting Regional Blood Volume

1. Tracking Spontaneous Arousal

- Head fixed mice
 - Full brain fUSI recordings
 - Neuropixel recordings
 - Visual cortex
 - Hippocampus
- Camera tracked spontaneous whisking
 - Whisking signal
- No stimulus was given
- Monitored blood volume across all regions (fUSI)
 - During whisking (start/stop)

2. Classifying Neurons by Arousal Response

- Labeled neurons from sim. Neuropixel recordings
 - Correlated firing rate ts. with whisking ts.
 - Arousal+ firing rate increased during whisking
 - Arousal- firing rate decreased during whisking
- Fit separate HRFs for each group of neurons
 - Averaged all neurons in each arousal group
 - Regressed each time series against fUSI blood volume
 - -> HRF weights
- Convolved set of HRF weights with each arousal time series
 - -> Blood volume timeseries
 - Correlated with original fUSI -> r value

3. Testing Across Sleep and Wake

- Mice spontaneously transitioned between sleep and wake
 - Compared to 2 group model (exp 2)
- Used the Arousal labels and HRF weights from Exp 2
 - Convolved each state (+)/(-) with HRF
 - Correlated with labeled fUSI time series
 - Compared predictive power

4. Mapping Population Distribution

- Mapped spatial distribution of Arousal+ / Arousal-
 - Across each cortical/subcortical region recorded
 - fUSI + Neuropixels dataset
- Used an IBL Neuropixels dataset (55 regions)
 - Correlation w/ whisking for each neuron in each region
 - Label each as Arousal+/- based on that correlation
 - Map those proportions across the whole brain

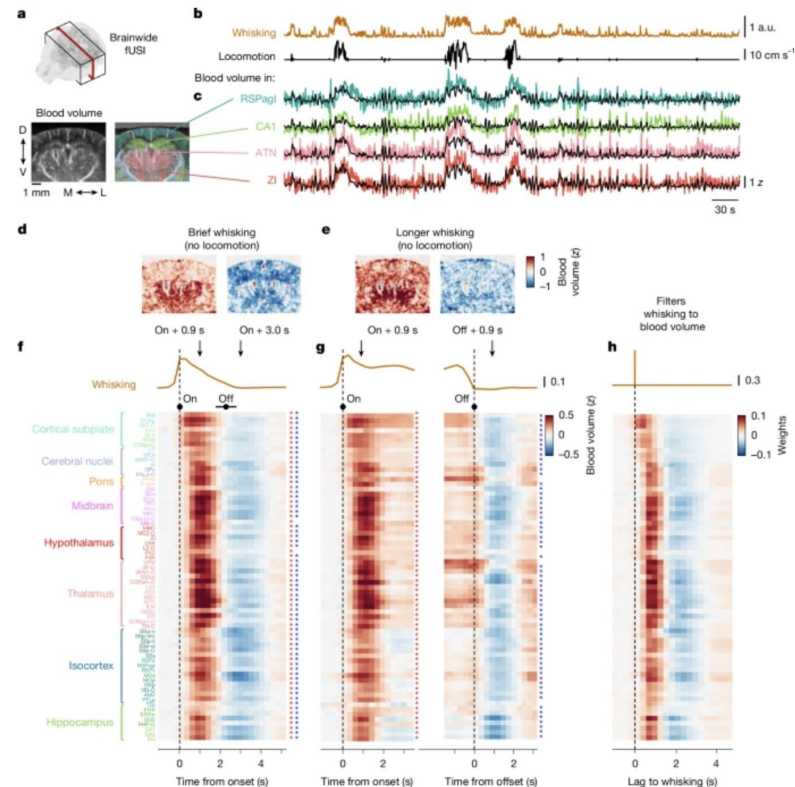
5. Predicting Regional Blood Volume

- Correlated the:
 - *Arousal+ / Arousal- ratio* to
 - *Magnitude of blood vol change during arousal (fUSI recording)*
 - Averaged all values where whisking was on
- Across all regions
 - fUSI & Neuropixels dataset
- Compared this to:
 - Bulk firing rate time series correlated to
 - Original fUSI recording for that region

1. Arousal events modulate brainwide blood volume

- Whisking onset drives synchronized increases in blood volume
 - Across the entire brain simultaneously
- Arousal events produce an entire hemodynamic response
 - Not a local one

Fig. 1: Arousal events modulate brainwide blood volume.



2. Firing of arousal-defined populations predicts blood volume

- Each group of neurons has a distinct HRF
 - Arousal+ peaks @ ~1.2s
 - Arousal- peaks @ ~2.1s
 - With opposite sign
- Separate modeling predicts blood volume better than bulk firing rate
 - $r = 0.63$ vs 0.44

Fig. 2: Bulk firing rate does not predict arousal-related changes in blood volume.

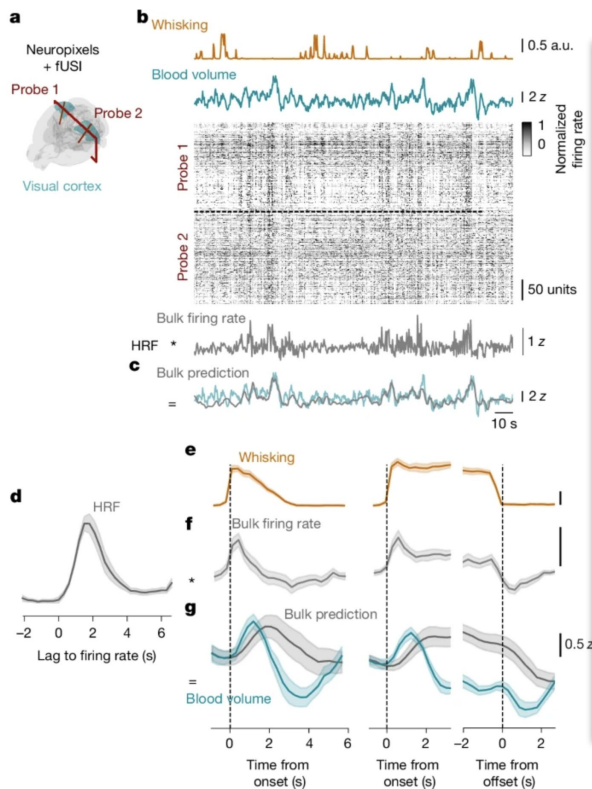
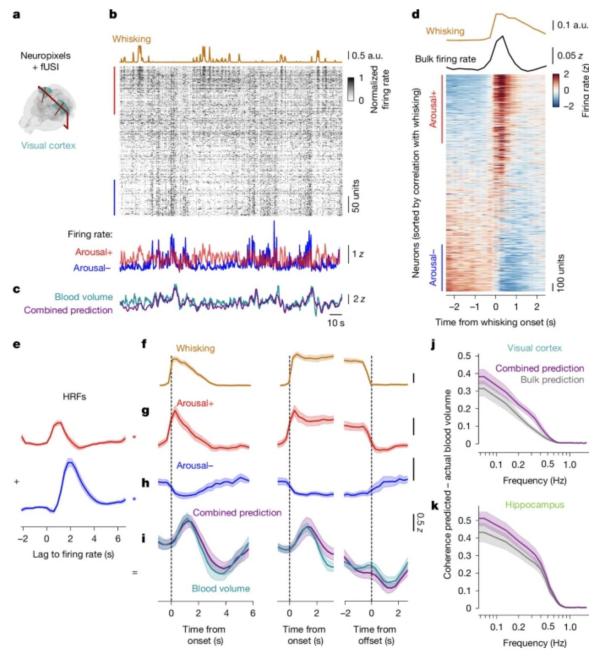


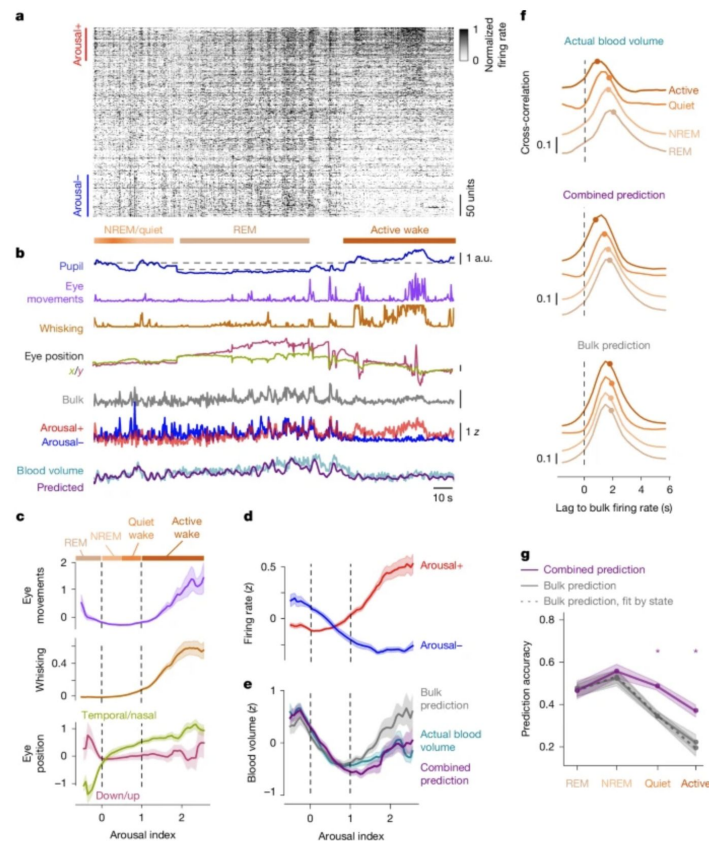
Fig. 3: The firing of two arousal-defined populations predicts blood volume.



3. The two populations account for blood volume across states

- Model predicts blood vol consistently across sleep/wake
- HRFs & population classification remain stable
 - Neurovascular coupling in the 2 neuron groups is state-independent

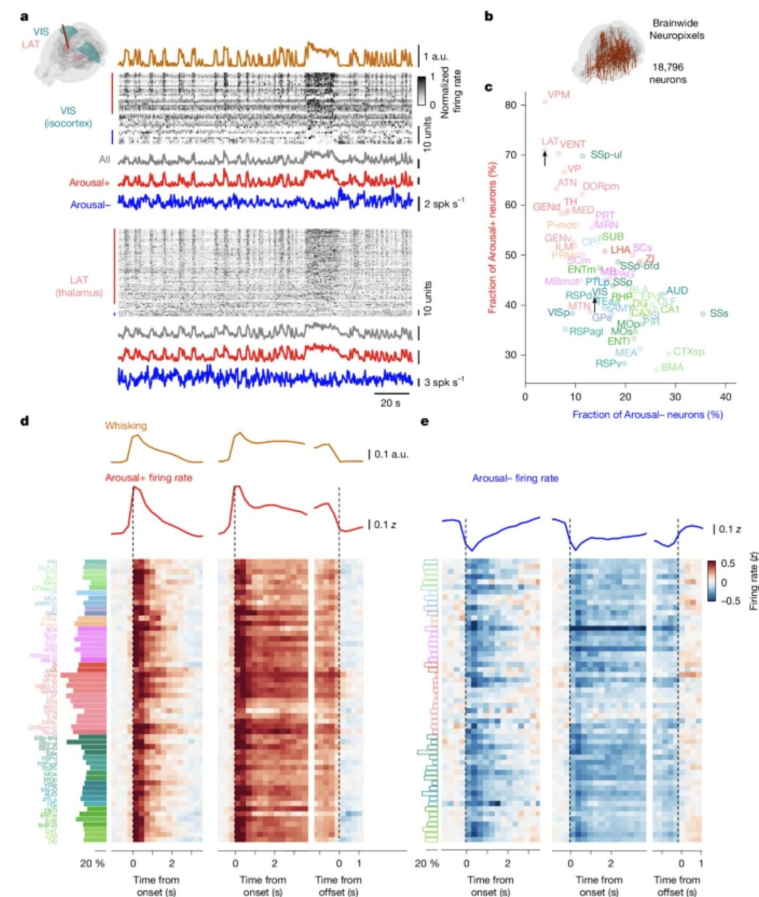
Fig. 4: The two populations account for blood volume across all states.



4. The two populations are present throughout the brain

- Both neuron groups coexist in every brain region
- Proportions vary per region

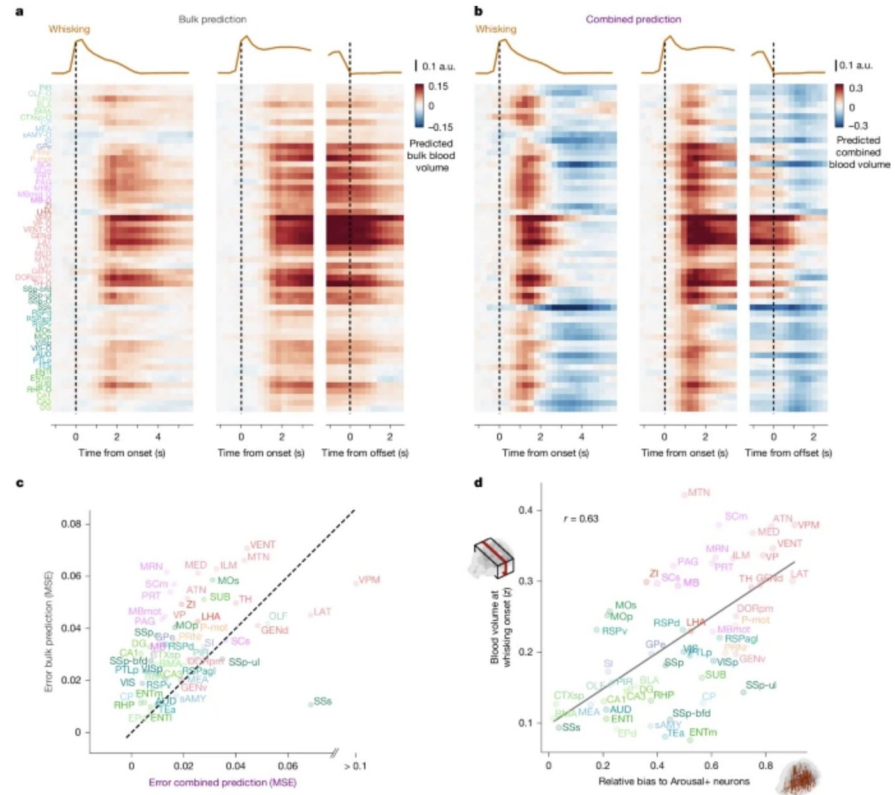
Fig. 5: The Arousal+ and Arousal- neuronal populations are present throughout the brain.



5. The two populations account for brainwide blood volume changes

- The population ratio explains regional differences in blood volume response
 - Better than bulk firing rate
- Regions aren't different because their neurovascular coupling works differently
 - differences depend on the amount of Arousal+ / Arousal- neurons

Fig. 6: The Arousal+ and Arousal- populations account for brainwide blood volume.



Conclusion

- Previously, if researchers saw a negative correlation between two brain regions, they assumed those two regions were hardwired into competing cognitive networks.
- The paper suggests that a negative correlation can appear simply because one region has a higher abundance of Arousal- neurons and the other has more Arousal+ neurons .
- The "network" behavior we see in human resting-state fMRI may actually just be a global map of our shifting arousal states.
- Two positives regions
 - Tracking global arousal together OR
 - Working in a shared functional task?

Takeaway

- Every fMRI voxel or ROI is already a mixture of Arousal+/- contributions
 - Which can't be separated directly
- But future experiments could mitigate these effects by adding a continuous arousal proxy (pupil dilation, heart rate, etc)
 - Regress out the arousal proxy
- Modalities that capture neural currents directly (EEG, MEG, etc) could in aggregate capture arousal +/- structure